LESSON 5.1

Determine the Area of Parallelograms and Rhombuses

Set B · Version A · Second form · supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can find the area of a parallelogram using base × height.

WHY THIS MATTERS Your architecture firm is designing a parallelogram-shaped patio for a client's backyard. The client needs to know how many square feet of pavers to order. The patio has a base of 14 feet and a height of 9 feet.

REMEMBER

Determine the Area of Parallelograms and Rhombuses

Area of Parallelograms & Rhombuses. Formula: Area = Base × Height (A = b · h) A parallelogram is a four-sided shape with two pairs of parallel sides, like a leaning rectangle. Its area is how much flat space is inside it.

1. Identify the base (any side).
2. Identify the perpendicular height (height must be perpendicular to base, not slanted).
3. Multiply base times height; label with square units (e.g. cm², in²)

Words: **Slanted side *(Lado inclinado)*** **Parallelogram *(Paralelogramo)*** **Parallel *(Paralelas)*** **Base *(Base)*** **Height *(Altura)***

1 **Circle the letter of the best answer. Show how you know.**

A bus travels 45 miles each hour. Which equation gives the distance d after h hours?

(A) d = 45h
(B) d = 45 + h
(C) d = h ÷ 45
(D) d = 45 − h

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

Using d = 45h, how far does the bus travel in 6 hours?

(A) 7.5 miles
(B) 45 miles
(C) 51 miles
(D) 270 miles

My work — one step per line	What I did to both sides

1. **Look** *What is done to the variable?*
2. **Undo** *Do the opposite — both sides.*
3. **Simplify** *One side at a time.*
4. **Check** *Put it back in.*

3 **Circle the letter of the best answer. Show how you know.**

A table shows x = 1, 2, 3 with y = 6, 12, 18. Which equation fits?

(A) y = x + 5
(B) y = 6x
(C) y = x + 6
(D) y = 6 + x

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

In c = 3n, the cost c depends on the number n. Which is the dependent variable?

(A) n
(B) 3
(C) c
(D) Both of them

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

A landscape architect is tiling a parallelogram-shaped courtyard. The courtyard has a base of 18 feet and a height of 10 feet. Each tile covers 2 square feet. What is the area of the parallelogram courtyard?

(A) 28 sq ft
(B) 56 sq ft
(C) 90 sq ft
(D) 180 sq ft

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

6 **Circle the letter of the best answer. Show how you know.**

A landscape architect is tiling a parallelogram-shaped courtyard. The courtyard has a base of 18 feet and a height of 10 feet. Each tile covers 2 square feet. How many tiles does the architect need?

(A) 45
(B) 90
(C) 180
(D) 360

SHOW YOUR WORK

EXPLAIN YOUR THINKING

If you cut a triangle from one side of the parallelogram and move it to the other side, what shape do you get? How does this help you find the area?

SENTENCE STARTER The area of the parallelogram equals base x height = ___ x ___ = ___ square feet, which is the same as a ___ with the same base and height.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

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PART 1 Understand the idea

models and meaning

1 **Write the name of the correct group on each line.**

Each label describes a segment on a parallelogram blueprint. Sort each one into 'Is the height' or 'Is NOT the height'.

Groups: Is the height | Is NOT the height

_____ A dashed line straight up from the base, making a 90 degree angle

_____ The slanted left edge of the parallelogram

_____ The perpendicular distance from the base to the top side

_____ The long tilted side that connects the bottom to the top

_____ The straight-up distance measured at a right angle to the base

_____ The leaning side that is longer than the perpendicular gap

2 **Write the letter of the matching item on each line.**

Match each parallelogram's base and height to its correct area (A = base x height).

_____ base 6 ft, height 7 ft

_____ base 13 ft, height 2 ft

_____ base 4 ft, height 11 ft

_____ base 15 ft, height 4 ft

(A) 42 sq ft

(B) 26 sq ft

(C) 44 sq ft

(D) 60 sq ft

PART 2 Apply it

real-world problems

3 **Circle the letter of the best answer. Show how you know.**

A landscape architect is tiling a parallelogram-shaped courtyard. The courtyard has a base of 18 feet and a height of 10 feet. Each tile covers 2 square feet. Why use the height instead of the slanted side length?

(A) Because the slanted side is always longer

(B) Because parallelograms have no height

(C) Because the base is on the bottom

(D) Because area needs the perpendicular distance between the two bases

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Cut the slanted end off a parallelogram and slide it to the other side. What shape forms, and why does that show A = base × height?

(A) A bigger parallelogram, so the area gets larger

(B) A triangle, so the area is cut in half

(C) A square, so all four sides must be equal

(D) A rectangle with the same base and height, so its area equals base x height

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

A blueprint parallelogram has a base of 8 ft, a height of 5 ft, and a slanted side of 7 ft. What is its area?

(A) 26 sq ft

(B) 35 sq ft

(C) 40 sq ft

(D) 56 sq ft

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

6 **Circle the letter of the best answer. Show how you know.**

Two parallelograms both have base 10 ft and height 6 ft. N is more tilted (slant 9 ft). Compare their areas.

(A) Blueprint N is larger because its slanted side is 9 ft

(B) Blueprint M is larger because it is less tilted

(C) Equal: both are 10 x 6 = 60 sq ft because area depends on base and height, not the slant

(D) You cannot compare them without the slanted side of M

SHOW YOUR WORK

EXPLAIN YOUR THINKING

If you cut a triangle from one side of the parallelogram and move it to the other side, what shape do you get? How does this help you find the area?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

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